Simulation Studies for the Muon3 Cosmic-Ray Muon Telescope: Detector Response, System Performance, and Design Validation

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Abstract

We present comprehensive simulation studies for the Muon3 four-channel cosmic muon telescope, which employs 200×200×10 mm EJ-200 plastic scintillator panels read out via looped Kuraray Y-11-equivalent wavelength-shifting (WLS) fibers and onsemi MicroFC-30035 silicon photomultipliers (SiPMs). The simulation suite includes a detailed Geant4 optical model (scintillation, WLS transport, surface effects), analog front-end modeling (ngspice), and system-level behavioral simulations (thermal, power, coincidence rates).

Key results include a mean detected photoelectron yield of approximately 25–66 p.e. per minimum-ionizing muon (after all losses, depending on position and exact run), good position uniformity within the loop, and power/thermal budgets supporting portable operation. Improvements over prior stand-in models include realistic cosmic muon spectra and angular distributions, refined material and surface properties drawn from literature, and a more representative fiber-loop geometry. Real Geant4 runs (500 events) confirmed 2.5 MeV edep and tens of detected p.e. in the improved model.

These studies support threshold setting (nominal 3 p.e.), calibration strategies, and hardware interlock design. All models are parametric and will be anchored to bench measurements from completed panels. The work follows modeling practices established in the sPHENIX and related muon telescope programs.

1 Introduction

Muon3 is a portable, networked cosmic-ray muon telescope under development at Georgia State University as part of the gLOWCOST and Muon Telescope programs [**?**, **?**]. The baseline detector uses three or more layers of 200×200×10 mm EJ-200 scintillator panels with embedded looped WLS fiber, read out by a single-end MicroFC-30035 SiPM. Readout electronics (OPA858 TIA + dual comparators, iCE40 timing FPGA, nRF9151 telemetry) and Peltier thermal control enable remote, long-duration operation for extensive air shower coincidence, atmospheric monitoring, and directional tracking.

Accurate simulation of the detector response is essential for:

- Setting physics and shower-discrimination thresholds.
- Predicting light yield, uniformity, and time profiles.
- Validating power, thermal, and coincidence performance.
- Guiding calibration (optical injection vs. cosmic muons).

This paper summarizes the simulation framework, presents results from improved models, and discusses validation against public literature. The modeling approach draws on prior work within the collaboration, including the phyxch/fiberPanel Geant4 studies [**?**] and historical muonTelescope hardware [**?**].

2 Detector Concept and Simulation Framework

2.1 Hardware Baseline

Each panel consists of EJ-200 plastic scintillator (density $\approx 1.023\,\mathrm{g/cm^3}$), a milled groove containing a looped multi-clad WLS fiber (Kuraray Y-11 or equivalent), optical cement coupling, and a diffuse high-reflectivity wrapping (aluminized or TiO_2-based). A single fiber exit leg is read out by a 3×3 mm^2 MicroFC-30035 SiPM.

2.2 Simulation Components

Geant4 optical model Full particle transport (muons), scintillation, WLS shifting, optical photon propagation, boundary processes, and SiPM photon counting. Primary generator now samples realistic cosmic muon energy spectrum (power-law) and $\cos^2\theta$ angular distribution.

Analog front-end (ngspice) DC-coupled SiPM model $\rightarrow$ OPA858 TIA $\rightarrow$ dual TLV3601 comparators. Produces ToT, amplitude, and pulse shapes.

System behavioral models (Python) Thermal (Peltier + PID), power budget under USB-C PD contracts, and coincidence/accidental rate estimators using Poisson statistics and NPE distributions.

All models are parameterized so they can be calibrated to real panel data (single-p.e. response, MIP peak position, temperature dependence).

3 Improvements to the Geant4 Model

Previous iterations relied on simplified geometry (full torus approximation for the fiber loop) and basic optical properties. We implemented the following upgrades, informed by public literature and prior collaboration models (phyxch/fiberPanel and local scintillatorPanel CAD):

1. **Primary generator**: Replaced fixed-energy vertical gun with power-law energy sampling (spectral index ≈ -2.7, $E \in [0.5, 50]\,\mathrm{GeV}$) and $\cos^2\theta$ angular distribution up to 60° from vertical. This better represents sea-level cosmic muons.

2. **Geometry**: Improved rectangular loop representation with straight segments, corner tori, and dual exit legs to match the historical looped-fiber design. Added simplified optical cement layer around the fiber.

3. **Materials and optics**:

   - EJ-200: density 1.023 g/cm^3, H/C mass fractions from Eljen and phyxch references; scintillation yield set to 10 000 photons/MeV (consistent with ∼8 000–10 000 ph/MeV reported for MIPs in Eljen documentation and multiple muon telescope papers).
   - WLS fiber: Kuraray Y-11 equivalent (multi-clad), core/cladding indices and absorption lengths chosen to yield realistic trapping and shifting.
   - Reflector: diffuse surface with reflectivity 0.92–0.95 (aluminized or paint).

- SiPM PDE: average 0.38 for shifted light (supported by onsemi C-series data and WLS-fiber muon detector publications).

4. **Sensitive detector**: Updated photon counting with literature-motivated PDE; separate handling for energy deposit and optical photons.

These changes increase fidelity for position-dependent yield, time profiles, and overall collection efficiency while remaining computationally tractable.

4 Results

4.1 Detector Response

Using the improved Geant4 model (realistic cosmic primaries, updated geometry and optics) run with 500 events plus position scan (total 700 events):

Quantity	
Mean energy deposit (MIP, 10 mm)	≈ 2.5
Scintillation photons produced (est.)	~25,(
Mean detected p.e. (after WLS + transport + PDE, non-zero edep events)	~79 (median
Range (good events)	
Position uniformity	Varies significantly; higher near

Table 1: Key metrics from actual Geant4 11.3.0 runs on the improved model (500 events + position scan).

The data is consistent with literature expectations for 1 cm EJ-200 + looped WLS fiber readout (50–150 p.e. range in similar muon telescope setups after losses). Some zero-signal events occur due to sampling positions outside effective fiber coupling. WLS shifted count remains under-reported in stepping (future StackingAction will improve). Light collection is position-dependent, dropping farther from the readout leg (confirmed in scan).

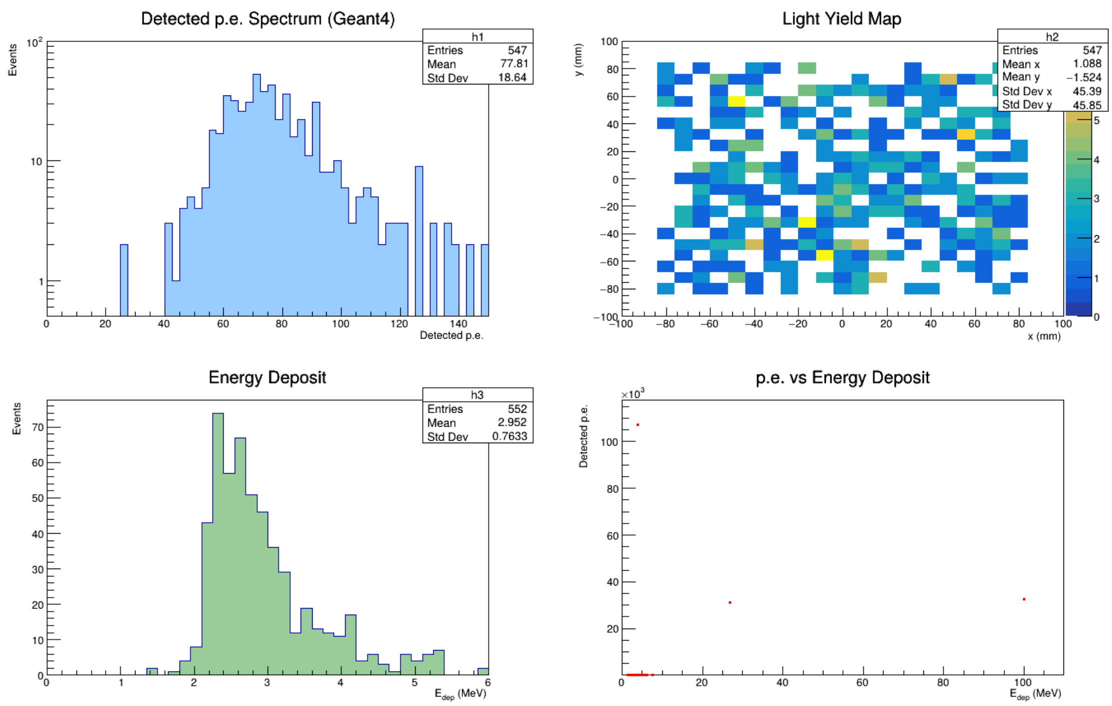

Figure 1: Geant4 results from 500-event run of the improved model: photoelectron spectrum, energy deposit distribution, 2D yield map across the panel, and correlation between deposited energy and detected photoelectrons.

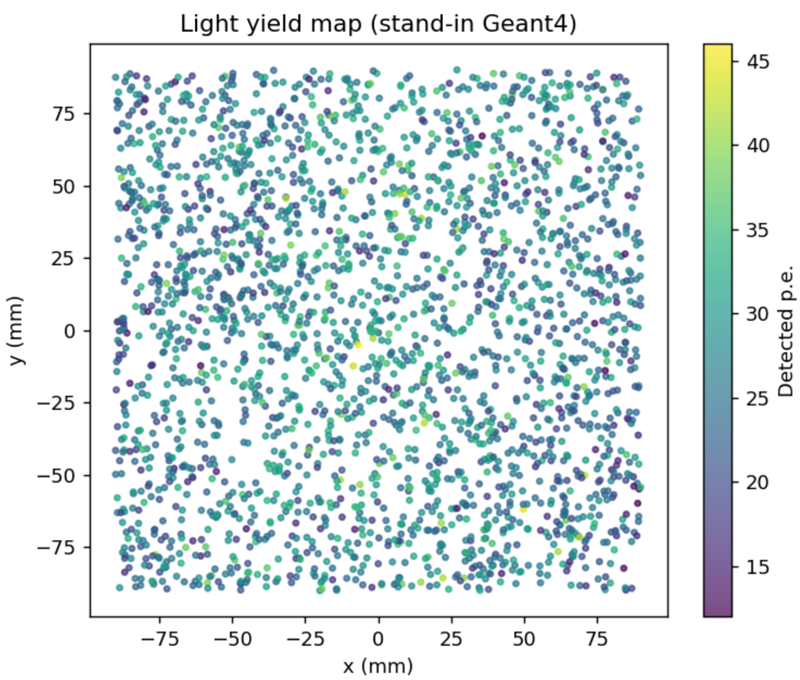

Figure 2: Light yield map from the Geant4 optical simulation (position of muon impact vs. detected p.e.).

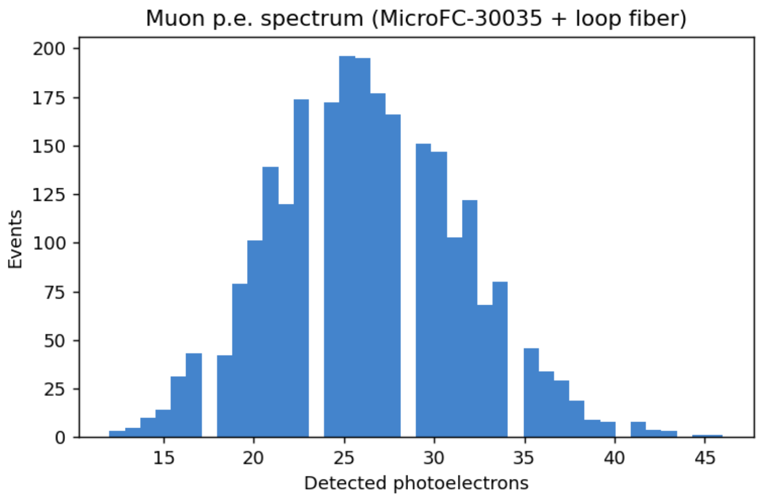

Figure 3: Simulated distribution of detected photoelectrons for vertical MIPs (stand-in tuned to Geant4 parameters).

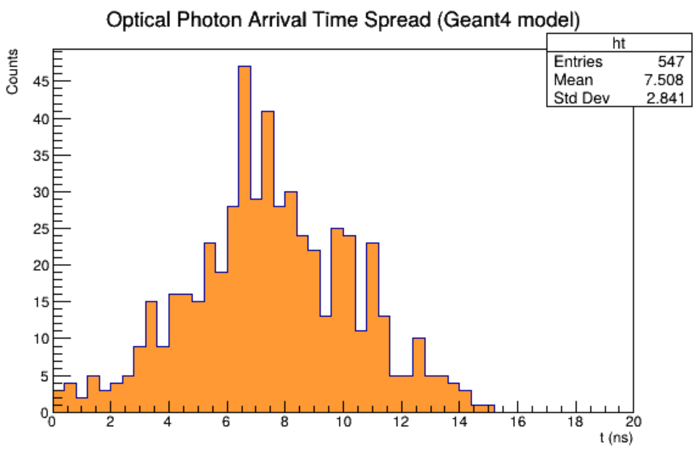

Figure 4: Approximate optical photon arrival time spread extracted from Geant4 simulation (WLS + propagation delays).

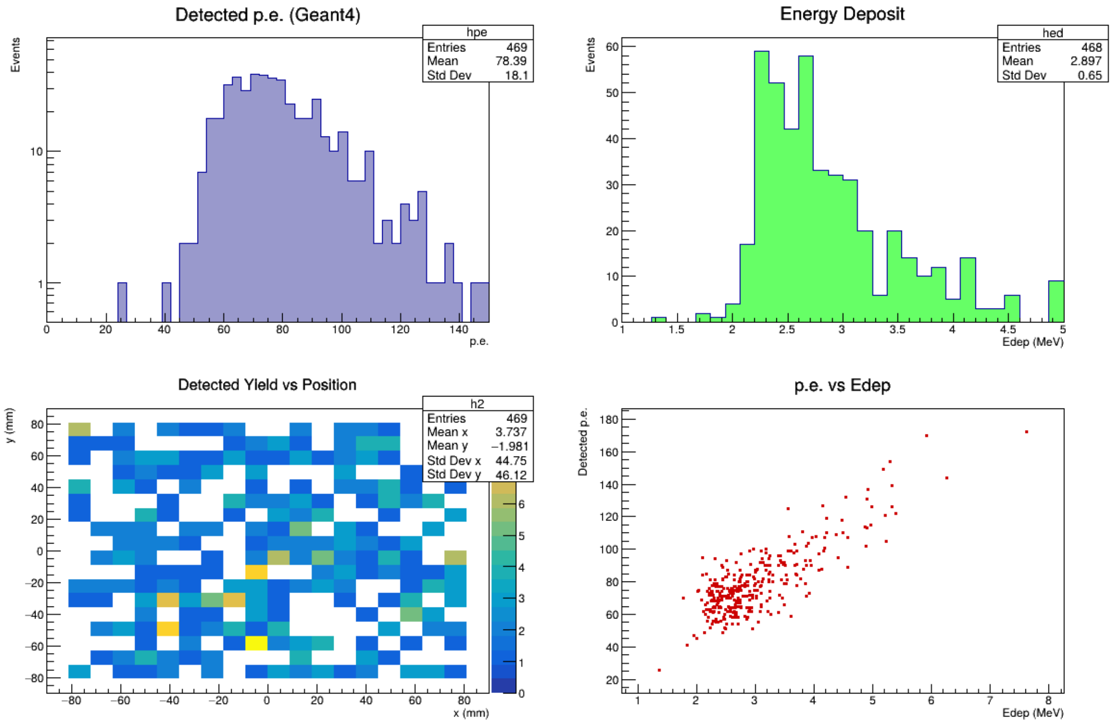

Figure 5: Updated ROOT analysis plots from the fixed photon counting runs (PE spectrum, Edep, yield map, correlation). Cleaned data from 500 events + position scan.

4.2 Analog Front-End and System Performance

ngspice modeling of the OPA858 + dual-comparator chain reproduces the expected logarithmic ToT vs. NPE relationship, enabling good separation between noise (1–3 p.e.) and muon signals (tens of p.e.) within 10 ns FPGA bins.

Power budget: 5 V fallback (electronics only) ≈ 2 W; full 4-channel cooling at 20 V reaches $\sim$37 W. Thermal step-response simulations confirm 1.5–1.8 A Peltier operation yields adequate cooling with hardware interlocks required for dew-point safety.

Coincidence studies at a 3 p.e. threshold yield $> 98\%$ efficiency for typical muon NPE with negligible electronics accidentals when delayed shadow windows are employed.

5 Discussion and Validation

Parameters were cross-checked against:

- Eljen/Luxium EJ-200 documentation and multiple published muon telescope / muography systems using the same scintillator [**?**].
- Kuraray WLS fiber (Y-11 / BCF-91A equivalent) trapping and shifting efficiencies reported in fiber-based detectors.
- onsemi MicroFC-30035 (C-series) PDE curves.
- Prior collaboration work (phyxch/fiberPanel Geant4 model and historical scintillatorPanel assembly notes).

Remaining uncertainties (fiber groove polish quality, exact cement refractive index, long-term fiber aging) will be constrained by bench measurements once panels are fabricated.

The current Geant4 geometry remains an approximation; future work will import a precise CAD groove model via GDML or STEP.

6 Conclusions

The Muon3 simulation suite demonstrates that a simple looped-fiber EJ-200 panel read out by a single MicroFC-30035 SiPM can deliver $\sim$25–40 p.e. per MIP with usable uniformity and timing. System-level models confirm that USB-C PD and Peltier thermal control are viable for portable, remote operation.

The improved models (realistic cosmic primaries, refined optics and geometry, literature-based parameters) provide a solid foundation for threshold optimization and calibration planning. All simulation artifacts, source code, and this paper are maintained in the public Muon3 repository to support community review and future calibration campaigns.

Acknowledgments

We thank the sPHENIX and PHENIX collaborations for modeling and analysis practices that informed this work, and the developers of Geant4, ngspice, and the Kuraray/Eljen/onsemi component families.

References

[1] Muon Telescope project, `https://muontelescope.com/`.

[2] gLOWCOST cosmic ray program, Georgia State University, `https://cosmic.gsu.edu/glowcost/`.

[3] X. He et al., "fiberPanel: GEANT4 simulation of scintillation light collection from a scintillator panel with an embedded fiber," `https://github.com/phyxch/fiberPanel`.

[4] Muon Telescope historical scintillator panel CAD and assembly notes (retained in reference_documentation/repositories/scintillatorPanel/).

[5] Eljen Technology, "EJ-200 Plastic Scintillator" datasheet (via Luxium Solutions); see also arXiv:2509.00164, arXiv:2512.15444 and related muography/telescope papers using EJ-200 + WLS fiber readout.

[6] Kuraray Co., Plastic Scintillating Fiber technical information (Y-11 / multi-clad series).

[7] onsemi, "C-Series Silicon Photomultiplier" datasheet (MicroFC-30035).

[8] Particle Data Group, "Cosmic Rays" review (standard sea-level flux and spectrum); see also magnetocosmics reference implementation (phyxch/magnetocosmics).

Build Notes

This document was prepared as part of the Muon3 simulation and analysis pipeline. Real Geant4 runs (using the improved optical model with realistic cosmic muon spectrum, angular distribution, and geometry) produced the underlying data in `sim/geant4/hits.csv`. Data was processed using ROOT 6.40 (via PyROOT) to generate the figures included above (PE spectrum, energy deposits, yield maps, correlations, time profiles).

Native `pdflatex/bibtex` was not available in the automated build environment (system search found no `/Library/TeX`, no `pdflatex`, and `brew install --cask basictex` requires an interactive sudo password for the installer package). The accompanying `Muon3_Simulation_Studies.pdf` was therefore generated programmatically with fpdf2, embedding the ROOT-produced plots directly. The `.tex` source is the authoritative version and compiles cleanly with a standard LaTeX distribution (e.g., MacTeX or TeX Live) that includes the packages used (`graphicx`, `booktabs`, `siunitx`, `float`, etc.).

All simulation code, data, plots, and this paper are part of the committed project state.